INCH-POUND

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DETAILED SPECIFICATION

HELMET COVER, REVERSIBLE, MARPAT™ WOODLAND AND DESERT PATTERN CAMOUFLAGE, MARINE CORPS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for the helmet covers that are worn over the Marine Corps Lightweight helmet, and the Personnel Armor Systems, Ground Troop (PASGT) helmet, and the Modular Integrated Communications Helmet (MICH).

1.2 Classification. The helmet cover assembly will be of one type and in the following sizes as specified (see 6.2).

Type I – Marine Corps Reversible Helmet Cover
Type II – Modular Integrated Communications Helmet (MICH) Reversible Helmet Cover

1.3 Schedule of sizes. The helmet cover will be constructed in the following sizes (see 6.2):

SCHEDULE OF SIZES

Type I	**Type II**
X-Small/Small	Medium/Large
Medium/Large	
X-Large	

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be address to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COET (Bldg 6), 700 Robbins Ave., Philadelphia, PA 19111-5092 or emailed to James.Heiman@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at www.dodssp.daps.mil.

AMSC N/A **FSC 8415**

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-4	- Glossary of Fabric Imperfections
FED-STD-595	- Colors Used in Government Procurement

COMMERCIAL ITEM DESCRIPTION

A-A-55126	- Fastener Tapes, Hook and Loop, Synthetic
A-A-50199	- Thread, Poly Core: Cotton-or Polyester-Covered

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075	- Label: For Clothing, Equipage, and Tentage, (General Use)
MIL-PRF-5038	- Tape, Textile and Webbing, Textile, Reinforcing, Nylon

(Copies of documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY RESEARCH, DEVELOPMENT AND ENGINEERING COMMAND, NATICK SOLDIER CENTER

MARPAT Woodland Pattern Green 474
MARPAT Woodland Pattern Khaki 475
MARPAT Woodland Pattern Coyote 476
MARPAT Woodland Pattern Black 477

MARPAT Desert Pattern Urban Tan 478
MARPAT Desert Pattern Light Tan 479
MARPAT Desert Pattern Highland 480
MARPAT Desert Pattern Light Coyote 481

(Copies of drawings are available from the U.S. Army Soldier and Biological Chemical Command, Natick Soldier Center, ATTN: AMSSB-RIP-TS(N), Natick, MA 01760.)

DEPARTMENT OF DEFENSE

DoD Standard Color Card for Sewing Thread

CODE OF FEDERAL REGULATIONS

Title 40, Part 798.4470 (Primary Dermal Irritation)

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(Copies of documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-737	- Air Permeability of Textile Fabrics
ASTM-D-3775	- Fabric Count of Woven Fabrics
ASTM-D-3776	- Mass Per Unit Area (Weight) of Fabric, Option C
ASTM-D-5034	- Breaking Force and Elongation of Textile Fabrics (Grab Test)
ASTM-D-6193	- Stitches and Seams

(Copies of documents are available on line at www.astm.org or from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC-8	- Colorfastness to Crocking: AATCC Crockmeter Method
AATCC-15	- Colorfastness to Perspiration
AATCC-16	- Colorfastness to Light
AATCC-20A	- Fiber Analysis: Quantitative
AATCC-81	- pH of the Water-Extract from Wet Processed Textiles
AATCC-61	- Colorfastness to laundering, Home and Commercial: Accelerated
AATCC-96	- Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC Evaluation Procedure 6	- Instrumental Color Measurement
AATCC Evaluation Procedure 9	- Visual Assessment of Color Differences of Textiles

(Copies of documents are available on line at www.aatcc.org or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASQC Z1.4	- Sampling Procedures and tables For Inspection by Attributes

(Copies of documents are available online at www.asq.org or from the American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

MISCELLANEOUS

Principle and Methods of Toxicology, A Wallace Hayes (editor), 1989, pages 394-396

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(Applications for copies of referenced documents should be addressed to Raven Press, 1185 Avenue of the Americans, New York, NY 10036.)

Marzulli, F. and H. Maibach, "Contact Allergy: Predictive Testing in Humans," Advances in Modern Toxicology, Volume 4, pages 353-372, 1977

(Applications for copies should be addressed to U.S. Army Center for Health Promotion and Preventive Medicine, ATTN: MCHB-DC-TTE, Bldg., E-2100, Aberdeen Proving Grounds, MD 21010-5422.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design. The helmet cover is reversible and can be worn in both desert and woodland environments. The Type I cover is designed to fit over the Marine Corps Lightweight Helmet, and the Personnel Armor System, Ground Troops (PASGT) helmet. The Type II cover is designed to fit over the Modular Integrated Communications Helmet (MICH), with front and back openings, and crown loop fastener tape to accommodate communications equipment. See Figures 1 and 2. The cover provides the helmet with signature reduction in the visible and near infrared spectral ranges, which provides against detection by visual observation or through passive night vision devices. The cover also provides noise reduction (softening noise if helmet strikes something). The cover contains buttonholes that allow for the attachment of indigenous plant material and other camouflage enhancing aids.

3.4 Basic material. The basic material for the reversible helmet cover shall be midweight nylon/cotton twill cloth printed with a desert pattern print on the face side and the woodland pattern print on the back side, and conforming to the requirements specified below. The cloth shall show no toxicity when used as intended and when tested as specified in 4.4.1 (see 6.2e). The suggested sole source shall be in accordance with 6.6.

3.4.1 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a nominal cut length of 1-1/2 inches and a round cross-section, with a nominal denier of 2.25 to 2.5. The use of any form of nylon waste is prohibited, such as undrawn fiber, mixtures of deniers, lusters or cross sections, and waste from any stage of fiber production: whether drawn, undrawn, or mixed or garneted fiber. The contractor shall submit the fiber producer's certification that each lot nylon staple used conforms to the requirements specified herein. Testing shall be as specified in paragraph 4.4.2.

3.4.2 Cotton. The cotton shall be carded. Testing shall be as specified in paragraph 4.4.2.

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3.4.3 Yarn. The warp and filling yarn shall be singles made from a blend of 50 (± 5) percent nylon with the remaining percentage cotton, based on the dry weight of the desized cloth when tested as specified in paragraph 4.4.2.

3.4.4 Weave. The weave shall be a 2/1 left-hand twill.

3.4.5 Fiber and fabric identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act. Each roll shall indicate cloth type. Testing shall be as specified in paragraph 4.4.2.

3.4.6 Physical requirements. The cloth shall conform to the requirements in Table I when tested as specified in paragraph 4.4.2.

TABLE I – Physical requirements.

Wt. Per sq. yd. ounces	Yarns per inch minimum		Breaking strength minimum		Tearing strength minimum		Air permeability maximum
	Warp	Filling	Warp Pounds	Filling Pounds	Warp Pounds	Filling Pounds	Cu. ft./min./sq. ft.
7.0 (min.)	86	54	200	125	11	8	25.0

3.4.7 Color. The cloth shall be dyed and printed on the backside of the fabric with the woodland colors. The use of pigments may be allowed. The cloth shall be dyed to a ground shade approximating Khaki 475. The Woodland camouflage pattern shall be obtained by roller or screen-printing using either three or four rollers or screens, as appropriate for the Green 474, Khaki 475, Coyote 476 and Black 477 areas of the pattern.

The cloth shall be dyed and printed on the face side with the desert colors. The cloth shall be dyed and printed with the warp effect side as the face. The cloth shall be dyed to a ground shade approximating Light Tan 479. The desert camouflage pattern shall be obtained by roller or screen-printing using either three or four rollers or screens, as appropriate for the Urban Tan 478, Light Tan 479, Highland 480 and Light Coyote 481 areas of the pattern. Resin bonded pigments are not permitted.

3.4.8 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid shall be chosen and applied such that the dyed cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.6. When no standard sample is available, the dyed cloth shall show no more than a slight trace of labile sulfur when tested as specified in 4.4.2.

3.4.9 Color matching

3.4.9.1 Visual matching. The color and appearance of the camouflage printed cloth shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, under filtered tungsten lamps that approximate artificial daylight D75 illuminant with a color temperature of 7500 (± 200) K with illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under horizon lamplight at 2300 (± 200) K.

3.4.9.2 Instrumental color matching. Each color of the camouflage printed cloth shall match the standard sample using AATCC Evaluation Procedure 6. A color difference greater than a $\Delta E_{CMC} = 1.0$ when using a $\Delta E_{CMC}(2:1)$ ratio (D_{65} /10°) units as compared to the standard sample, shall be basis for rejection.

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3.4.10 Colorfastness. The finished camouflage printed cloth shall show fastness to: light (after 40 AATCC standard fading hours or 170 Kilojoules); laundering (after 4 cycles); and perspiration (acid and alkaline) and crocking. The colorfastness of the cloth shall be equal to or better than the standard sample, or equal to or better than a rating of "4" using the AATCC Gray Scale for Color Change and a rating of "3-4" using the AATCC Gray Scale for Staining for each of the colors, except the Black 477 shall have an equal to or better than a rating of 2-3. The woodland pattern print shall have a colorfastness to crocking rating when using the AATCC Chromatic Transference Scale Rating equal to or better than 3.0 for all colors except Black 477, which shall have a rating not lower then, 1.0 when tested as specified in 4.4.2.

3.4.11 Pattern execution. The pattern shall reproduce the standard sample in respect to design, colors, and registration of the respective areas. The pattern repeat, of the camouflage printed finished cloth shall be 36 (± 1.5) inches. Each pattern shall show solid coverage; skitteriness exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution, a pattern drawing will be provided, and the pattern of the finished cloth shall match that of the drawing.

3.4.12 Spectral reflectance. The desert pattern print finished cloth shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, 700 to 860 nanometers (nm) for the colors specified in Table II, when tested as specified in 4.4.2.1.

TABLE II – Spectral reflectance requirements.

Wavelengths Nanometers	Reflectance values (percent)					
	Light Tan 479		Light Coyote 481 & Highland 480		Urban Tan 478	
	Min.	Max.	Min.	Max.	Min.	Max.
700	38	53	19	41	25	44
720	38	54	20	41	25	45
740	39	55	20	42	25	46
760	40	56	21	42	26	47
780	41	57	21	42	27	48
800	43	58	22	43	28	50
820	45	59	23	45	30	52
840	48	62	24	46	33	55
860	50	65	25	48	36	58

3.4.13 Dimensional stability. The shrinkage or elongation both in warp and filling direction of the finished cloth shall be not greater than 3.5 for individual sample unit and not greater than 3.0 percent for the lot average when tested as specified in 4.4.2.

3.4.14 pH. The pH value of the water extract of the finished cloth shall be no lower than 5.0 or higher than 8.5 when tested as specified in 4.4.2.

3.4.15 Toxicity. The finished cloth shall not present a dermal health hazard when used as intended when tested as specified in 4.4.1.

3.4.16 Ground shade/printed seconds/mill seconds. Ground shade cloth shall be dyed in conformance with the specified basic material and shall meet the physical, mechanical, and dimensional requirements of the respective finished fabric. Printed seconds shall be defined as cloth that has been rejected only for defects pertaining to color, infrared reflectance, or camouflage print patterns, which are cited in the specified basic material requirements. Mill

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seconds finished firsts may contain slubs, knots (see 4.4.4) and FED-STD-4 for all other fabric defects, which constitute seconds.

3.4.16.1 Disposal of ground shade/printed seconds/mill seconds/rejected helmet covers. Non-usable seconds shall be disposed in one of the following manners:

1) Seconds in lengths of 1 yard or less may be disposed by using the contractor's practice.
2) Contact the USMC to determine if there is an alternate use for rejected fabric or finished helmet covers.
3) If the USMC does not identify any alternate use, the contractor may sell the fabric or helmet covers on the secondary market.

Rejected helmet covers shall be labeled or indelibly marked as "seconds/rejected by USMC".

3.4.17 Tape, nylon. The nylon tape for Type I shall conform to Type III, 3/4-inch wide of MIL-PRF-5038. The color of the tape shall be Camouflage Green 483. The cut and finish lengths (inches) of the tape shall be as follows:

	Small			Medium/Large			Large		
	Front	**Back**	**Side**	**Front**	**Back**	**Side**	**Front**	**Back**	**Side**
Cut Length (inches)	5	5-3/4	6-3/4	5	5	6-3/4	5	6-1/4	7-1/4
Finish Length (inches)	4-1/4	5	6	4-1/4	4-1/4	6	4-1/4	5-1/2	6-1/2

3.4.18 Fastener tape, hook and pile. For Types I and II the hook and pile fastener tape for the attachment straps shall be 3/4-inch wide and conform to Type II, Class 1 of A-A-55126. The color of the tape shall be Camouflage Green 483. For Type II, the hook and loop fastener tape for the communication flap closure shall be 5/8-inch wide and shall conform to Type II, Class 1 of A-A-55126. The color of the tape for the communication flap closure shall be Camouflage Green 483 for the woodland side, and shall be Khaki P-1 (C.A. 66019 or approximating color chip 30277 of FED-STD-595) for desert side. For Type II, the crown pile fastener tape shall finish 2 inches wide by 2 inches long. The color of the tape for the crown pile fastener tape shall be camouflage green 483 for the woodland side and shall be khaki P-1 (C.A. 66019 or approximating color chip 30277 of FED-STD-595) for the desert side.

3.4.19 Thread. The thread for the needle and bobbin (looper) shall be commercial size Ticket No. 50 (Tex size 36 to 50) with a minimum breaking strength of 3.2 lbs. when tested as specified. As an alternate, Ticket No. 70 (Tex size 31 to 35) with a minimum breaking strength of 2.6 pounds may be used as the bobbin (looper) thread. The thread shall conform to A-A-50199. All thread shall be non-staining and show good colorfastness to laundering when tested according to AATCC-61, Test 3A (4 cycles). The thread color shall be Camouflage Green 483 (approximating color chip 34094 of FED-STD-595) for the woodland side of the cover and Khaki P-1 (C.A. 66019 or approximating color chip 30277 of FED-STD-595) for the desert side of the cover.

3.5 Labels. Each helmet cover shall have the following legible label, which shall be printed or sewn onto the desert side of the facing band of the helmet cover.

The Type I label shall be as follows:

HELMET COVER, REVERSIBLE, MARINE CORPS
MARPAT™ WOODLAND/DESERT
CONTRACT #: SIZE:
NSN:
CONTRACTOR:

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The Type II label shall be as follows:

HELMET COVER, REVERSIBLE, MARINE CORPS
MARPAT™ WOODLAND/DESERT, MICH
CONTRACT #: SIZE:
NSN:
CONTRACTOR:

3.5.1 Label/tag. Each item shall be individually bar-coded with the Type VIII, Class 17 label/tag of MIL-DTL-32075. This label/tag shall be located so that it is completely visible on each item and causes no damage to the item. The tags shall have a hole and shall be attached to each item by a fastener. The tags shall be clearly legible and readable by a scanner. The bar-coding element shall be a 13 digit national stock number (NSN). There shall be a 12-digit UPC number assigned for each NSN by the Government. The initial "UPC" must appear beneath the code. The bar-code for NSN and UPC printing shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and shall cause no damage to the item. The UPC code must also be placed on all shipping containers on which the NSN appears.

3.6 Patterns. Patterns, which provide a 1/2-inch allowance for seaming side lining to crown, 3/8-inch allowance for all other seams, will be furnished to the contractor. The pattern list in Table III is provided to insure that the pattern set is complete.

TABLE III – Pattern parts.

Material	Nomenclature	Pattern Abbreviation	Cut Parts
Basic Material	Type I		
	Side Crown	CROWN_SIDE	2
	Center Crown	CROWN_CENTER	1
	Front Facing	FACING_FRONT	2
	Back Facing	FACING_BACK	2
	Type II		
	Side Crown	CROWN_SIDE	2
	Center Crown	CROWN_CENTER	1
	Front Facing	FACING_FRONT	2
	Back Facing	FACING_BACK	2
	Front Opening	FR_OP	1
	Back Opening	BK_OP	1
	Communications Flap	CMM_FLP	2
	Tab Template	HOOK_PILE	N/A

3.7 Configuration. The following specifics are needed in the reversible helmet cover to provide uniform appearance, comfort and durability in combat and operations other than war. End item construction and appearance shall conform to the requirements of this document and the finished dimensions as specified in Table VIa, VIb, and Figures 1 and 2 to maintain item configuration and compliance to end item and component performance tests (See 4.4).

3.7.1 Seaming. Seaming shall be consistent, exhibit a uniform appearance and shall conform to ASTM D-6193 stitch types listed below. All seams shall be flat with no protruding seam allowance or raw edges to create irritation or discomfort. To maintain durability and functionality the seams shall be sewn with 10-14 stitches per inch for all outside visible stitching. Overedge or pre-hemming shall be 6-10 stitches per inch. Material edges must not ravel. Raw edges may be turned-in, or turned-under to prevent raveling. The buttonholes in the cover crown shall be straight-cut, tacked at each end with a minimum of four crossover stitches, with a

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finished cut length of 3/4-inch. The buttonholes in the cover facing shall be straight cut or eyelet-end tapered bar type buttonholes, with a finished length of 1-1/4 to 1-3/8 inches. Seaming shall be as specified in Table IVa and IVa.

TABLE IVa – Seaming (Type I).

Seam placement	Seam type	Gage	Stitch type
Join side crown to center crown	LSc-2	1/4-inch	301
Join front and back facings to crown	SSe-2		301
Join front facing to back facing	SSa-1 1/		301
Attach loop fastener tape to nylon tape on both sides	Box-X		301
Attach nylon tape to facing	Box-X		301

1/ SSa-1 with stitch class 500 (503, 504 or 505)

TABLE IVb – Seaming (Type II).

Seam placement	Seam type	Gage	Stitch type
Stitch communication flap	SSe-2		301
Baste pleats in communications flap (according to pattern)	SSa-1		301
Attach communication flap and back opening to center crown	LSb-1		301
Stitch back opening	SSc-1		
Bartack communications flap with 1-inch bartack	Bartack		301
Attach front opening to center crown	LSb-1		301
Stitch front opening	SSc-1		301
Join side crown to center crown	LSc-2	1/4-inch	301
Join front and back facings to crown	Sse-2		301
Join front facing to back facing	SSa-1 1/		301
Hook and pile tab (Tab shall be stitched such that the hook tape overlaps the loop tape.)	Lsa-1		301
Attach hook and loop fastener tape to front and back facing with a 1/4-inch box-X stitch	Box-X		301

1/ SSa-1 with stitch class 500 (503, 504 or 505)

3.8 Finished measurements. The finished measurements shall conform to Tables Va and Vb. The measurements of the helmet cover shall be taken with the cover folded in half and laid flat with the bottom edges even. (See Figures 1 and 2).

TABLE Va – Finished Dimensions (Type I).

Size	½ Circumference 1/	Height 2/	Tolerance
X-Small/Small	16-3/4 inches	9-1/4 inches	+ 1/2, -0
Medium/Large	17-1/2 inches	9-1/2 inches	+ 1/2, -0
X-Large	18-1/4 inches	10-1/2 inches	+ 1/2, - 0

TABLE Vb – Finished Dimensions (Type II).

Size	½ Circumference 1/	Height 2/	Tolerance
Medium/Large	17 inches	9-1/2 inches	+ 1/2, -0

1/ The circumference measurements shall be taken from the back center of the center crown panel along the contour of the helmet to the front center of the center crown. (See Figures 1 and 2.)

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2/ The height measurement shall be taken from the top mid point of the center crown to the midpoint of the side crown bottom edge. (See Figures 1 and 2.)

3.9 Figures. Figures are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall control.

3.10 Workmanship. The finished cover shall conform to the quality of product established by this document, and the occurrence of defects shall not exceed the specified quality levels.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.3), it shall be examined for the defects specified in 4.4.2, 4.4.3 and 4.4.4. Failure of any test or inspection shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANZI/ASQC Z1.4. The lot size shall be expressed in units of helmet covers. The sample unit shall be one helmet cover. The inspection level shall be S-2 in accordance with ANSI/ASQC Z1.4.

4.4 Component and end item inspections. In accordance with 4.1, components and end item shall be tested in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable procurement documents. The government reserves the right to inspect all components and end items to determine conformance to requirements.

4.4.1 Toxicity assessment. The contractor must furnish information, which certifies that the finished product is composed of materials, which have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact. A certificate of conformance is required. The Government reserves the right to inspect such items to determine the validity of the certification.

4.4.2 Material and end item testing. Material and end item testing shall be performed in accordance with the test methods specified in Table VI. The cloth shall be tested for the characteristics listed in Table VI. All tests reports shall contain the individual values utilized in expressing the final results. For material testing the sample unit shall be five continuous yards full width of the finished cloth, for all physical and chemical tests. The lot shall be considered unacceptable if one or more sample units fail to meet any requirements specified.

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TABLE VI – Material Testing.

Characteristics	Requirement Paragraph	Test Method
Material		1/
Nylon	3.4.1	
Identification		1/
Luster		1/
Denier		1/
Absence of nylon waste		1/
Cotton	3.4.2	
Identification		1/
Fiber Content	3.4.3	
Cotton content		AATCC 20A 1/, 2/
Nylon content		AATCC 20A 1/
Weave	3.4.4	Visual 1/
Weight	3.4.6	ASTM D-3776
Break Strength	3.4.6	ASTM D-5034
Tear Strength	3.4.6	ASTM D-3775
Yarns per Inch	3.4.6	1/
Air Permeability	3.4.6	ASTM D-737
Presence of Labile Sulfur	3.4.8	1/
Visual Color Matching	3.4.9.1	AATCC Evaluation Procedure 6
Instrumental Color Matching	3.4.9.2	AATCC Evaluation Procedure 9
Colorfastness		
Light (after 40 hrs or 170 kilojoules)	3.4.10	AATCC 16 A or E
Laundering (after 4 cycles)	3.4.10	AATCC 61 test 1A
Crocking	3.4.10	AATCC 8
Perspiration (acid and alkaline)	3.4.10	AATCC 15
Spectral Reflectance	3.4.12	4.4.2.1
Dimensional Stability	3.4.13	AATCC 96 Test Vic. A 3/
pH	3.4.14	AATCC 81
Toxicity	3.4.15	4.4.1

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The cotton content shall be calculated as follows:

Cotton content, percent = R/S x 100

R = Weight of residual fibers

S = Weight of dry desized specimen

3/ Except that 1993 AATCC Standard Reference Detergent (non-phosphate) without optical brightener shall be used.

4.4.2.1 Spectral reflectance. Spectral reflectance data shall be determined on the desert side and shall be obtained from 700 to 860 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the barium sulfate, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g. magnesium oxide or vitrolite tiles. The spectral bandwidth shall be less than 26 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CIE source A or CIE source D65. The specimen shall be measured as a single layer, backed with six layers of the same fabric and shade. Measurements shall be taken on a minimum of two different areas and the data averaged. The measured areas should be taken at least 6 inches away from the selvage. The specimen shall be

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viewed at an angle no greater than 10 degrees from the normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall be 0.3725 inches in diameter. Any color having spectral reflectance values outside the limits at four or more of the wavelengths specified shall be considered a test failure.

4.4.2.2 Color matching.

4.4.2.2.1 Visual color matching. The color and appearance of the cloth shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, under filtered tungsten lamps that approximate artificial daylight D75 illuminant with a color temperature of 7500 (+ 200) K with illumination of 100 (+ 20) foot candles, and shall be a good match to the standard sample under horizon lamplight at 2300 (+ 200) K.

4.4.2.2.2 Instrumental color matching. Each color of the camouflage printed and finished cloth shall match the standard sample using AATCC Evaluation Procedure 6. A color difference greater than a $\Delta E_{CMC} = 1.0$ when using a ΔE_{CMC}(2:1) ratio (D_{65} /10°) units as compared to the standard sample, shall be basis for rejection.

4.4.3 Finished dimensions. The helmet cover shall conform to the dimensions listed in Table VIa and VIb.

4.4.4 End item visual examination. The reversible helmet covers shall be examined for the defects listed below. All fabric defects shall be scored in accordance with Table VII, which are clearly noticeable at normal viewing distance (3 feet), or known to and effect serviceability and appearance of the garment. Fabric defects are defined in Section I of FED-STD-4.

TABLE VII – End item visual defects.

Examination	Defect	Classification	
		Major	Minor
Cloth	Any hole, cut, or tear	101	
	Any broken or missing yarns or multiple floats	102	
	Any mend, darn, or patch	103	
	Any needle chew	104	
	Any visible mend	105	
	Any run, thin place, dye streak, misweave, knot or slub affecting appearance or serviceability. 1/		201
	Any permanent fold, pleat, or crease affecting appearance or serviceability.		202
	Knots greater than Sears Scale Level C.	106	
	Slubs greater than Sears Scale Level D.	107	
	Not as specified	108	
Components and assembly	Any component caught in any unrelated operation of stitching (not otherwise classified herein)	109	
	Any component part omitted or not attached as specified (not otherwise classified herein)	110	
	Any exposed drill hole	111	

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TABLE VII - End item visual defects. - Continued

Examination	Defect	Classification	
		Major	Minor
Fastener tape	Omitted, not specified type or color	112	
	Any hole, cut or tear	113	
	Hooks crushed affecting closure	114	
	Not as specified	115	
Tape, nylon	Any hole, cut, or tear	116	
	Multiple floats, smash, or burn	117	
	Not as specified	118	
Seams and stitching:			
Open seams	Up to and including 1/2-inch		203
	More than 1/2-inch	119	
	NOTE: A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		
Raw edges (except where required or otherwise classified herein)	More than1/4-inch when securely caught in stitching		204
	NOTE: Raw edges not securely caught in stitching shall be classified as open seams.		
Seam and stitch type	Wrong seam or stitch type	120	
	Seam pleated or excessively puckered, distorted, pleated, wavy, or twisted, clearly noticeable		205
Stitch tension	Tension loose, resulting in loose bobbin or top thread		206
	Tension excessively tight, resulting in puckering of material		207
Rows of stitching	Any row omitted	121	
	Topstitching sewn too close to edge resulting in damage to cloth.	122	
Stitches per inch (all types of stitching)	One stitch less than minimum specified		208
	Two or more stitches less than minimum specified	123	
	One or more stitches in excess of maximum specified		209
	NOTE: Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows: - Within the minor defect classification – no defect - Within the major defect classification – minor defect		

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TABLE VII - End item visual defects. - Continued

Examination	Defect	Classification	
		Major	Minor
Thread breaks, skipped stitches or run-offs	Overstitched less than 3/4-inch in each direction beyond the defective stitching area. NOTE: On all types of stitching, thread breaks or two or more consecutive skipped or run-off stitches not overstitched shall be classified as open seams.		210
Stitching ends (on type 301 stitching)	Ends of stitching not secured as specified (except when caught in other stitching or turned under in a hem)		211
Buttonholes	On cover crown - Up to two omitted - Three or more omitted On cover facing - One or more omitted Stitches not securely caught in fabric, causing stitches to pull away from fabric One or more broken stitches or two or more skipped stitches Tack on one or both ends omitted Uncut Not specified type	 124 125 126	 212 213 214 215 216
Shade	Shade does not meet approved shade samples, which fall within the shade band of the standard sample. Thread/cloth color not as specified.	127 128	
Cleanness	Spot, stain, excessive thread ends not trimmed or removed, odor, affecting appearance or serviceability.	129	
Label	Omitted, incorrect, illegible, not as specified, or not attached where specified; Attachment causes damage to the item. Ends of material not seared Searing forms sharp edges. Color not as specified.	 130 131	217 218 219
Barcode	Omitted or not readable by scanner Human readable interpretation (HRI) omitted Not visible on folded, packaged item. Causes damage to the end item	 132	220 221 222

1/ As defined in FED-STD-4 Glossary of Fabric Imperfections.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be a specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain

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requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reversible helmet cover is for wear by military personnel of the United States Marine Corps for use in garrison and combat missions.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Types, classes and sizes required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article is required (see 3.1, 4.2 and 6.3).
e. Approval is required prior to purchase of this fabric (3.4).
f. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Fabric defect scales. Fabric defect replica kits are available from:

Sears Roebuck and Company
Department 817 (ATTN: BSC 23-29)
Sears Tower Chicago IL 60684

6.6 Approved cloth source. The only Marine Corp approved source for the cloth is:

Bradford Dyeing Association
P.O. Box 539
Westerly, RI 02891

6.7 Subject term (key word) listing.

Uniform
Clothing
Desert
Utility

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6.8 National Stock Number (NSN).

Type I
8415-01-500-3903
8415-01-500-3904
8415-01-500-3905

Type II
8415-01-506-9503

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - MC

Preparing activity:
DLA - CT

Project Number
8415-0267

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at www.dodssp.daps.mil.

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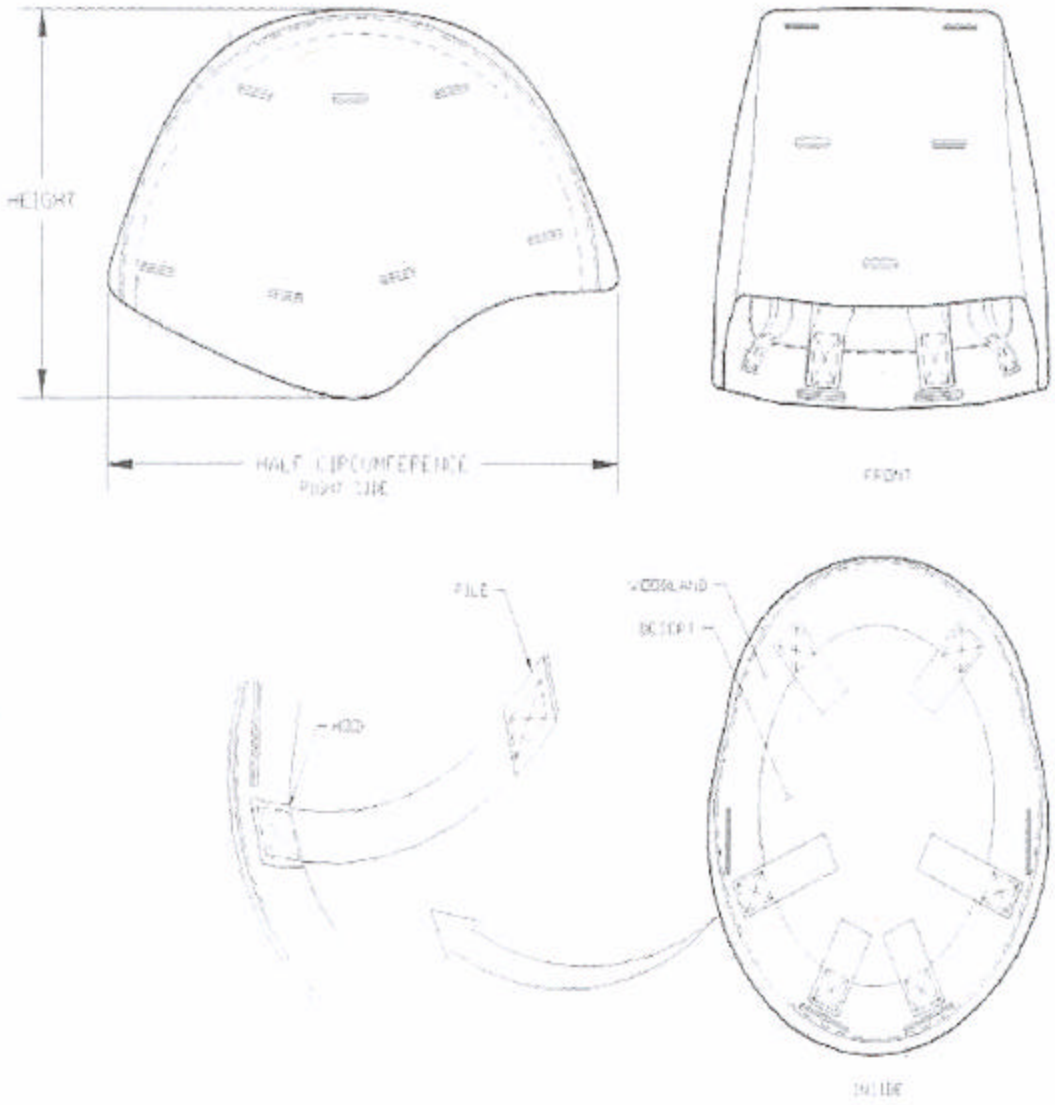


Figure 1. (Type I) – PASGT (Full) Cut Helmet with Web Suspension

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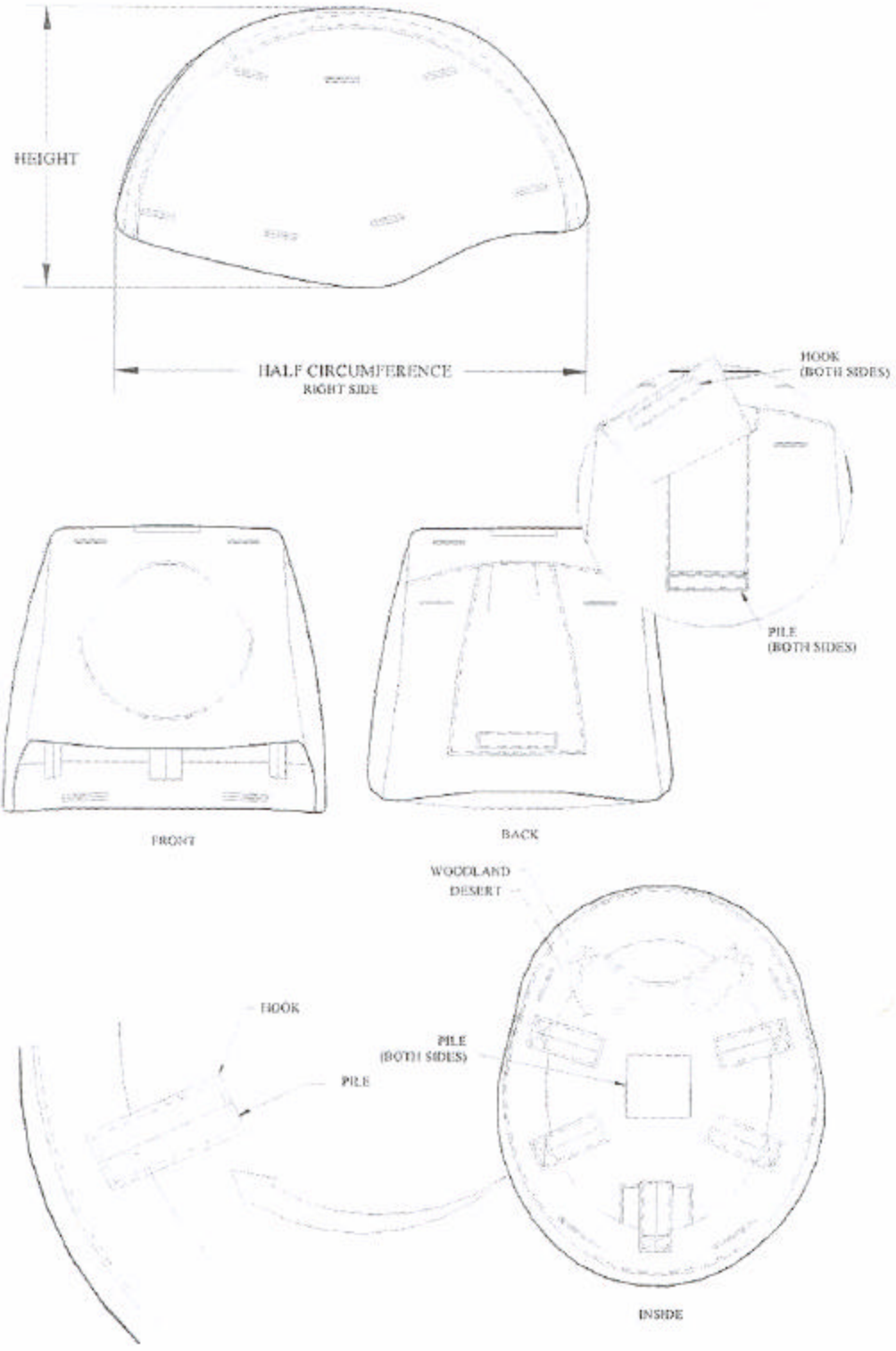


Figure 2. (Type II) – Tactical Cut Helmet with Pad Suspension

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